Week 5: Coordinated & Spread Spectrum Access

5.1 Introduction to Multiple Access Techniques

Wireless communication networks must enable a multitude of independent user terminals to share a common, bounded physical medium—the unguided radio frequency spectrum—efficiently, fairly, and with minimal cross-talk. As established in Week 4, the open airwaves constitute a shared broadcast domain where uncoordinated simultaneous transmissions mix destructively, causing packet collisions Tanenbaum and Wetherall (2010). While Multiplexing (Week 3) describes the static or base-level partitioning of a composite channel for multiple underlying data streams (such as splitting an LTE channel into parallel subcarriers), Multiple Access strategies define the dynamic Link-Layer protocols and MAC sublayer mechanisms that allow independent user nodes to actively request, coordinate, and utilize those subchannels Tanenbaum and Wetherall (2010), Molisch (2022). These schemes are fundamentally divided into Coordinated/Contention-Free frameworks (where a central orchestrator or rigid physical geometry enforces orthogonal user separation) and Spread Spectrum frameworks (where robustness is achieved by expanding the signal bandwidth) Molisch (2022).

5.2 Structural Breakdown of Coordinated Multiple Access Techniques

Coordinated access methodologies rely on strict deterministic slicing of physical resources to ensure that multi-user transmissions remain mathematically orthogonal to one another, completely eliminating intra-cell multi-user collisions Molisch (2022).

5.2.1 Frequency Division Multiple Access (FDMA)

FDMA segments the total allocated frequency band into separate, non-overlapping narrowband frequency subchannels. Each active mobile terminal is granted exclusive ownership of a specific subband for the entire duration of its connection session.

a. **Physical Structure & Orthogonality:** Users transmit simultaneously, but their signals are separated entirely in the frequency domain. Because physical radio frequency filters cannot achieve a mathematically perfect rectangular passband, real-world hardware deployments must separate adjacent channels using unallocated spectrum buffers known as Guard Bands to prevent Adjacent Channel Interference (ACI) Tanenbaum and Wetherall (2010).

b. **Performance Profiles:**

- **Advantages:** Minimal synchronization complexity at the terminal side; supports continuous, unbroken transmission streams without buffer-and-burst latency.
- **Limitations:** Highly inefficient under bursty data traffic (the channel remains idle if the assigned user is silent); rigid bandwidth granularity limits dynamic scaling; highly vulnerable to frequency-selective fading where a deep localized fade can completely drop an individual user's subchannel Molisch (2022).
- **Real-World Context:** Served as the baseline physical layer architecture for analog first-generation (1G AMPS) cellular networks.

5.2.2 Time Division Multiple Access (TDMA)

TDMA allocates the entire width of a single radio frequency channel to multiple users by dividing the transmission timeline into cyclical, recurring time frames partitioned into dedicated Time Slots.

a. **Physical Structure & Orthogonality:** Each user is synchronized to transmit exclusively within their assigned periodic time slot and must remain completely silent during all other slots. Because

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user terminals are scattered at varying spatial distances from the base station (e.g., across different zones of the Kirkuk University campus), their signals experience different propagation delays. To prevent the trailing edge of one user's packet from colliding with the leading edge of the next user's packet, TDMA frames must incorporate brief padding intervals known as Guard Times Molisch (2022).

b. **Performance Profiles:**

- **Advantages:** Highly flexible bandwidth allocation (a single node can be dynamically assigned multiple slots to increase data rates); effectively accommodates bursty digital traffic; simpler radio hardware since transmitters can turn off during inactive slots, saving battery power.
- **Challenges:** Mandates strict, highly complex master-clock time synchronization across all roaming terminals; system throughput can degrade due to multipath delay spreads if the arrival delay exceeds the guard time boundaries Tanenbaum and Wetherall (2010).
- **Real-World Context:**Power-efficient infrastructure behind digital second-generation (2G GSM) cellular systems.

5.2.3 Code Division Multiple Access (CDMA)

CDMA completely abandons the physical partitioning of time and frequency. All active user terminals transmit simultaneously and continuously across the exact same wideband frequency spectrum.

a. **Physical Structure & Orthogonality:** Users are separated mathematically by assigning a unique, highly specific binary spreading code sequence to each terminal. To separate these overlapping signals at the base station receiver, the codes must exhibit low cross-correlation properties Dordal (2023). When orthogonal codes (such as Walsh-Hadamard sequences) are utilized on a perfectly synchronized downlink, the mathematical inner product of two differing codes is exactly zero, enabling complete signal isolation Du and Swamy (2026).

b. **The Critical Power Control Constraint:** As noted in Week 4, CDMA paths are bound by the Near-Far Problem. Because all users share identical spectrum blocks, a terminal operating near the central base station will project an immense received power signature that can easily submerge the faint, heavily attenuated signal arriving from an edge node. This disrupts code orthogonality and induces massive Multi-User Interference (MAI). Thus, CDMA systems mandate ultra-fast, strict closed-loop power control circuits to continually scale terminal outputs, ensuring all user arrivals match identical power thresholds at the base station antenna array Du and Swamy (2026).

c. **Soft Capacity Dynamics:** Unlike FDMA and TDMA, which possess hard boundaries restricted by the absolute count of hardware channels or time slots, CDMA exhibits a Soft Capacity Limit. The network does not issue busy signals; instead, adding a new user marginally raises the background noise floor for all existing users until the cumulative interference reaches the decoding threshold of the base station receiver Molisch (2022).

- **Real-World Context:** Formed the technological backbone of third-generation (3G UMTS/WCDMA) networks.

5.3 Spread Spectrum Principles

Spread spectrum techniques intentionally modulate a narrow baseband information signal onto a transmission channel whose physical bandwidth is orders of magnitude wider than the minimum bandwidth mathematically required by the data payload (Molisch, 2022). This deliberate spectral inflation alters the signal's vulnerability profile to achieve three distinct engineering advantages:

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1. **Jamming Immunity:** Narrowband interference or intentional jamming signals only disrupt a tiny fraction of the expanded signal power, rendering the transmission robust against harsh interference.

2. **Low Probability of Intercept (LPI):** By spreading a fixed transmission power over a massive bandwidth, the signal's spectral density drops below the environmental thermal noise floor, causing the signal to appear as harmless background noise to unauthorized interceptors Du and Swamy (2026).

3. **Multipath Rayleigh Fading Resilience:** When a signal's spread bandwidth exceeds the coherence bandwidth of the radio channel, the frequency-selective fading drops affect only portions of the signal. Furthermore, in the time domain, DSSS allows receivers to implement RAKE architectures—specialized parallel correlator branches that isolate distinct multipath echo rays, shifting destructive multi-wave interference into constructive diversity gain Molisch (2022).

The core performance metric of any spread spectrum architecture is its Processing Gain (Gp), which mathematically quantifies the system's inherent noise-rejection capability:
$Gp = Bss/Bd = Ts/Tc$ Where:
Bss is the spread spectrum bandwidth,
Bd is the original baseband data bandwidth,
Ts is the baseband symbol duration, and
Tc is the underlying chip duration Molisch (2022).

5.3.1 Direct Sequence Spread Spectrum (DSSS)

In a DSSS architecture, the raw narrow-rate binary data stream is combined directly with a high-rate pseudo-random binary sequence known as a chipping code.

a. **Modulation Mechanics:** Each individual data bit is combined with a multi-bit pseudo-random noise (PN) sequence (typically via an explicit logical XOR operation) prior to carrier modulation Tanenbaum and Wetherall (2010), Dordal (2023). The duration of a single bit within this code sequence is defined as a Chip. Because the chip duration (Tc) is drastically shorter than the data symbol duration (Ts), the resulting output signal acquires an ultra-wide spectral footprint Dordal (2023).

b. Receiver Despreading: The receiver passes the wideband incoming signal through a localized correlator matched precisely to the identical PN sequence. This operation collapses the target wideband signal back into its original narrowband structure. Simultaneously, any narrowband interference or jamming encountered along the airwaves is multiplied by the PN code and spread across a massive bandwidth, drastically reducing its power spectral density and neutralizing its impact on the data stream Dordal (2023).

- **Code Requirements:** DSSS requires pseudorandom code families (such as Barker codes, Gold codes, or Kasami sequences) that demonstrate excellent mathematical autocorrelation properties to facilitate fast clock synchronization and lock tracking Du and Swamy (2026).
- **Real-World Context:** Utilized in global positioning systems (GPS) and legacy Wi-Fi standards (IEEE 802.11b) Dordal (2023).

5.3.2 Frequency Hopping Spread Spectrum (FHSS)

FHSS achieves its wideband footprint not by continuously expanding the signal simultaneously, but by rapidly changing the carrier frequency of a standard narrowband signal over time.

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a. Operational Mechanics: The total allocated spectrum block is partitioned into a large set of discrete frequency channels. The transmitter and receiver change their active operating frequency in a synchronized fashion according to a pseudo-random hopping pattern sequence Tanenbaum and Wetherall (2010). The duration for which the transmitter remains anchored to a single frequency step is defined as the Dwell Time.

b. **Classification Profiles:**

- **Slow Frequency Hopping (SFH):** The hopping rate is slower than the modulation symbol rate. Multiple consecutive data symbols are transmitted completely on the same frequency step before a hop occurs Molisch (2022).
- **Fast Frequency Hopping (FFH):** The hopping rate is faster than the modulation symbol rate. The carrier frequency shifts multiple times during the lifespan of a single data symbol, providing exceptional protection against frequency-selective fading and tracking jammers Du and Swamy (2026).
- **Real-World Context:** Power-conscious, low-complexity hardware baseline powering Bluetooth WPAN standards Tanenbaum and Wetherall (2010).

5.4 Direct Architectural & Physical Comparison

To summarize the structural parameters driving Layer 2 resource regulation, Table 1 describes the physical properties of each individual access model:

Table 1: Structural Property Configurations of Multiple Access Methods

Technical Parameter	**FDMA**	**TDMA**	**CDMA**	**DSSS**	**FHSS**
Separation Domain	Frequency Subbands	Time Slots	Orthogonal Codes	Continuous Code Spreading	Discontinuous Carrier Hopping
Orthogonality Vector	Non-overlapping spectrum slices	Non-overlapping time frames	Zero-inner-product code sequences	High-frequency noise-like chipping	Time-synchronized frequency hopping
Hardware Boundary Requirement	Guard Bands (prevents filter bleeding)	Guard Times (absorbs propagation shifts)	Strict Closed-Loop Power Control	High-speed clock tracking loops	Agile frequency synthesizers
Primary Technical Constraint	Inflexible with bursty traffic	High synchro-nization overhead	Bounded by the Near-Far problem	Complex wideband tracking	Synchronized pattern alignment
Real-World Protocol Example	1G AMPS Cellular	2G GSM Cellular Systems	3G UMTS / WCDMA Networks	GPS; Wi-Fi 802.11b	Bluetooth WPANs

5.5 Evolving System Architectures: The Shift to 4G/5G OFDMA

As we studied in Week 2, modern telecommunication landscapes underwent a massive paradigm shift during the transition from 3G to 4G LTE and 5G New Radio (NR). While CDMA was the uncontested

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foundation of 3G, modern high-speed cellular networks completely abandoned CDMA-based spreading mechanics Peterson and Sunay (2020).

At ultra-wide bandwidths (20 MHz to 100 MHz+), CDMA systems suffer from catastrophic multi-user interference because severe multipath propagation delay spreads destroy code orthogonality. To solve this, modern networks shifted entirely to Orthogonal Frequency Division Multiple Access (OFDMA) Peterson and Sunay (2020), Molisch (2022). As established in Week 3, OFDMA splits a massive wideband channel into thousands of parallel, closely packed narrowband orthogonal subcarriers. Multiple access is achieved by dynamically scheduling multi-user coordinate blocks across specific subcarriers (frequency domain) and specific OFDM symbols (time domain). This eliminates intra-cell interference without the complex hardware burden of CDMA power control loops Peterson and Sunay (2020).

5.6 Synchronization, Acquisition, and Tracking

Because spread spectrum systems operate close to or below the ambient noise floor, receiver decoding circuits cannot extract data until they achieve perfect time alignment with the incoming signal's code phase or hopping pattern. This synchronization process is split into two sequential phases:

1. **Acquisition (Coarse Alignment):** A search phase where the receiver continuously shifts the phase of its local code generator relative to the incoming signal, calculating correlation values until a massive energy peak is detected, establishing a rough timing lock Du and Swamy (2026).
2. **Tracking (Fine Tuning):** Once a lock is acquired, active feedback loops (such as Delay-Locked Loops or DLLs in DSSS) continuously track micro-fluctuations caused by Doppler shifts, hardware clock drift, or physical terminal mobility to maintain tight sub-chip timing alignment Molisch (2022).

5.7 Multiuser Environments and Interference Management

In asynchronous or non-ideal wireless environments, multipath reflections alter code arrival times, causing orthogonal code sequences to lose their zero-inner-product properties. This introduces cross-talk known as Multi-User Interference (MAI), which degrades receiver sensitivity. To maximize multi-user capacity under these harsh conditions, modern base stations deploy three distinct mitigation strategies:

- **Successive Interference Cancellation (SIC):** The receiver decodes the strongest user's signal first, reconstructs its corresponding physical waveform, and mathematically subtracts it from the composite received signal, allowing weaker edge-node signals to be decoded cleanly Molisch (2022).
- **Code Design Optimization:** Restricting code selection strictly to specialized mathematical families like Gold codes or Kasami sequences, which maintain deterministic, ultra-low cross-correlation side-lobes even under asynchronous arrival conditions Du and Swamy (2026).
- Advanced Multi-Antenna Filtering: Leveraging the Beamforming and Spatial Multiplexing capabilities of Massive MIMO configurations (Week 3) to isolate users geographically, suppressing interference by directing narrow energy beams to specific coordinates Molisch (2022).

5.8 Channel Coding and Error Control

To defend against deep channel fades and bursts of noise, modern access methods pair physical-layer modulation with Forward Error Correction (FEC) and Interleaving. By applying advanced linear block codes, low-density parity-check (LDPC) codes, or convolutional turbo codes, the system injects

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structured algebraic redundancy into the bitstream. When combined with block interleaving (which scatters sequential bits across different time slots or frequencies), a localized burst fade along the airwaves does not destroy a whole packet. Instead, it appears as random, isolated bit errors that can be mathematically corrected by the receiver without forcing a costly MAC-layer retransmission Molisch (2022), Peterson and Sunay (2020).

5.9 Conclusion

- Coordinated access methods establish contention-free environments by separating users into distinct, orthogonal domains: FDMA splits the frequency spectrum, while TDMA divides time slots.
- CDMA breaks away from physical slicing by using unique spreading codes, giving it a soft capacity limit that is bound by multi-user interference rather than rigid hardware blocks.
- Spread spectrum systems (DSSS and FHSS) provide exceptional resilience against intentional jamming, frequency-selective fading, and interception by expanding the transmission bandwidth well beyond the baseband requirement.
- Driven by high data demands, modern wireless networks have transitioned from CDMA-based architectures to highly scalable, multi-carrier OFDMA frameworks in 4G LTE and 5G NR systems.

References

Dordal, P. L. (2023). *An Introduction to Computer Networks*. Loyola University Chicago Department of Computer Science. Release 2.0.11, Open-Access (CC BY-NC-ND), ISBN: N/A.

Du, K.-L. and Swamy, M. N. S. (2026). *Wireless Communication Systems*. Springer.

Molisch, A. F. (2022). *Wireless Communications: From Fundamentals to Beyond 5G*. Wiley-IEEE Press, 3rd edition. ISBN: 978-1-119-11721-6.

Peterson, L. and Sunay, O. (2020). *5G Mobile Networks: A Systems Approach*. Morgan & Claypool. ISBN: 9781681738888.

Tanenbaum, A. S. and Wetherall, D. J. (2010). *Computer Networks*. Pearson, 5th edition.

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